

The 8-dimensional octonions are the final, esoteric members of the family of *real division algebras*. The 2-dimensional complex numbers and the 4-dimensional quaternions form the amiable relatives. While complex numbers had been studied since the early Renaissance to solve polynomial equations, their systematic theory sprouted in the 19th century parallel to their applications in understanding light as waves. Absent the language of abstract vectors but requiring a tool to understand the 3-dimensional world, Hamilton (1843) extended these to quaternions. So effective was he that to this day, quaternions remain a boon for graphics and robotics.

Complex numbers model rotations in the plane, which quaternions extend to space. This invites intricacies; the additional rotational axes mean the order of rotations matters. Algebraically, $xy = yx$ holds for complex numbers but not quaternions. Cayley and Graves (1840s) discovered the octonions wherein the deficit of quaternions degenerates further with $(xy)z = x(yz)$ unsatisfied. Devoid of immediate natural application, octonions find uses in modern string theory, underlying nebulous mathematics.

Division algebras cause plentiful sporadic phenomena linking algebra with geometry. My dissertation asks if these exist in *motivic homotopy theory*. An archetypal example of these phenomena is that no other division algebras exist. Cayley's method gives an alien 16-dimensional algebra which forces dissolution of the very notions of division and length, but *a priori*, new algebras could be found via other methods.

Nonexistence is proved via *topology*, the study of shapes through the coarse lens of *homotopy* where we freely bend, shrink, and stretch shapes to discard minutiae. For example, cylinders shrink into circles. *Homotopy theory* aims to understand quantities, such as the number of holes in a shape, which persist beyond homotopy. Another ubiquitous quantity is the count of (topological) *vector bundles* on a shape, formed by attaching (hyper)planes at each point. The circle has 2. Attaching lines at each point, we can form a cylinder, or twist lines to form a Möbius strip.

Spheres (points in n -dimensional space at distance one from the origin) are fundamental building blocks of all topological objects. Thus, functions between spheres govern the core structure of topology. Understanding them is an elusive, central problem. We obtain spheres from our algebras by considering length one vectors, with their multiplication producing functions between spheres called *Hopf maps*. These deeply constrain vector bundles, and underlie *projective spaces*, classical shapes parameterizing lines for each algebra.

Adams and Atiyah (1966) used powerful abstract homotopical and sophisticated geometrical machinery to turn these constraints into a simple divisibility problem: when does 2^n divide $3^n - 1$? There are exactly three solutions, implying there are exactly 3 division algebras.

Hopf maps give projective spaces *cellular structures* by building them from spheres. Projective spaces exhibit rich symmetries via *algebraic groups*, collections of geometric symmetries. Octonionic projective space is extraordinary. The degenerate structure allows only an octonionic projective line and plane. Furthermore, algebraic groups essentially fall into 4 infinite families alongside 5 exceptional groups. Freudenthal (1958) proves these 5 all come from octonions.

Motivic homotopy theory, founded in the 1990s, continues this story in *algebraic geometry*, the study of *varieties*. These are shapes given by solutions to polynomial equations, such as conic sections. With finitely many solutions, polynomials (thus varieties) are rigid, yet homotopy necessitates flexibility. This tension between topology and algebra requires encoding topological data in abstract "spaces" alongside varieties. Consequently, the simplest problems in topology become difficult conjectures in algebraic geometry.

A central questions, then, is when abstract homotopical structure can be represented by concrete functions between varieties? Asok, Hoyois, and Wendt (2016) provide *affine representability* answering this, then prove cases of the *Bass-Quillen* conjecture (1973) to showcase motivic homotopy's power in answering classical questions. Since Serre (1955), (algebraic) vector bundles on varieties have motivated and driven modern geometry. Paraphrasing, Bass asks if these are preserved under homotopy, exactly the complicated algebraic version of an easy topological fact.

I study varieties *over the integers*, $\mathbb{Z}$: integer solutions to polynomial equations. These are universal; the integers lie inside the real and complex numbers and we can "transport" varieties through this inclusion. Concretely, $\mathbb{Z}$ encodes not just integer solutions, but solutions in all algebras, allowing one to prove exceedingly general yet still powerful theorems.

Consequently, motivic homotopy theory contains more information than classical homotopy theory. The aforementioned tension necessitates *motivic spheres* varying in both "algebraic" and "topological" dimension. Adding topological information may cause spheres to not be varieties. When they are is unresolved. Asok, Doran, and Fasel (2015) give partial results exhibiting certain *quadrics* (general conic sections) as motivic spheres recovering the information of classical spheres. They decompose motivic spheres into two complicated "hemispheres" and understand them homotopically. I utilized a motivic version of the Hopf map to understand them algebraically, constructing an explicit vector bundle over this "hemisphere" and addressing a question of Asok and Østvær (2019).

I then consider varieties giving motivic analogues of projective spaces, particularly the *motivic octonionic projective plane* OP^2 . Classically, it is built from two spheres via a Hopf map. I will show OP^2 admits a *motivic cellular structure*, a similar decomposition. To use affine representability, I will show OP^2 exhibits a exceptional symmetry shared by the topological plane by using work of Garibaldi, Petersson, and Racine (2022) on algebraic groups over $\mathbb{Z}$.

The motivic setting creates difficulties, but Panin and Walter (2010) overcome these for quaternionic space by adding vector bundles to the decomposition. I have proved the analogy suddenly breaks with OP^2 . Transporting this variety to the complex numbers gives us a topological vector bundle; another classical divisibility argument concludes this structure exists only in certain dimensions, with dimension 8 excluded. This is unexpected. The theory is intimately tied to vector bundles. As such, the homotopical phenomena one encounters almost always arise through them. Affine representability implies some variety exhibits the homotopical behavior of OP^2 and the Hopf map. Yet, it cannot be a vector bundle and is truly exotic. It remains to find this variety and to understand why we are forced to depart from the realm of vector bundles.